A.L.O.N.E. v2 - Comprehensive Product Requirements Document

Document Purpose: Provide a startup-grade product and engineering blueprint for transforming A.L.O.N.E. from a voice-enabled AI assistant into a humanized autonomous virtual executive assistant.

1. Executive Summary

A.L.O.N.E. is a local-first autonomous desktop assistant designed to understand users, manage tasks, execute actions, learn habits, and proactively assist with daily work. The long-term goal is to create a digital executive assistant capable of operating a computer, understanding context, maintaining long-term memory, and managing workflows.

2. Product Vision

Create an AI assistant that:
• Understands the user deeply
• Remembers long-term context
• Executes real-world actions
• Learns habits and preferences
• Operates proactively
• Preserves privacy through local-first architecture

3. Target Users

Primary Users:
• Solo founders
• Developers
• Freelancers
• Students
• Content creators

Secondary Users:
• Startup teams
• Knowledge workers
• Researchers

4. Core Product Pillars

1. Persistent Memory
2. Autonomous Actions
3. Proactive Assistance
4. Human-like Context Awareness
5. Privacy & Security
6. Learning and Adaptation

5. System Architecture

Voice Layer
Brain Layer
Memory Layer
Planning Layer
Execution Layer
Screen Intelligence Layer
Learning Layer
Safety Layer

6. Human Memory Architecture

Memory Categories:

- Identity Memory
- Preferences Memory
- Relationship Memory
- Project Memory
- Goal Memory
- Episodic Memory
- Workflow Memory

Success Criteria:
Assistant remembers user information across months.

7. Planning Architecture

Goal → Planner → Task Breakdown → Tool Execution → Verification → Feedback Loop

Capabilities:

- Multi-step reasoning
- Dynamic replanning
- Failure recovery
- Progress tracking

8. Screen Intelligence

Features:
• OCR
• Screenshot understanding
• IDE awareness
• Terminal awareness
• UI element detection
• Document understanding

Outcome:
Assistant can understand what is visible on the screen.

9. Proactive Intelligence

Features:
• Deadline monitoring
• Calendar monitoring
• Email monitoring
• Project progress monitoring
• Habit tracking

Outcome:
Assistant initiates conversations and reminders.

10. Safety Framework

Risk Levels:
LOW - Immediate execution
MEDIUM - Single confirmation
HIGH - Double confirmation
CRITICAL - Manual approval required

11. Phase-Wise Implementation Roadmap

Phase 1 (Month 1-2) - Human Memory Foundation

- Build structured user profile system
- Implement preferences memory
- Implement project memory
- Implement goals database
- Implement relationship memory
- Create memory retrieval service
- Add memory editing UI

Success Criteria: Success: Assistant remembers user, projects, goals, and relationships.

Phase 2 (Month 2-3) - Personal Operating System

- Task management engine
- Calendar integration
- Meeting tracking
- Deadline management
- Project dashboards
- Persistent notes

Success Criteria: Success: Assistant manages work and projects.

Phase 3 (Month 3-4) - Screen Intelligence

- Screenshot capture service
- OCR pipeline
- Screen understanding model
- IDE parser
- Terminal parser
- Error diagnosis engine

Success Criteria: Success: Assistant understands the user's screen.

Phase 4 (Month 4-6) - Autonomous Planning Agent

- Task planner
- Goal decomposition
- Verification engine
- Recovery engine
- Execution tracking
- Multi-step workflows

Success Criteria: Success: Assistant executes complex tasks independently.

Phase 5 (Month 6-8) - Proactive Intelligence

- Background monitoring
- Reminder engine
- Email awareness
- Calendar awareness
- Project awareness
- Deadline awareness

Success Criteria: Success: Assistant proactively helps users.

Phase 6 (Month 8-10) - Habit Learning

- Behavior tracking

- Workflow detection
- Pattern recognition
- Automation recommendations
- Custom workflow generation

Success Criteria: Success: Assistant learns user habits.

Phase 7 (Month 10-12) - Humanization Layer

- Emotional context engine
- Relationship awareness
- Long-term conversational memory
- Adaptive communication style
- Personalized recommendations

Success Criteria: Success: Assistant feels personal and contextual.

12. MVP Definition

Mandatory Features:
• Voice interaction
• Long-term memory
• Email automation
• WhatsApp automation
• Project tracking
• Goal tracking
• Screen intelligence
• Multi-step planning

13. Success Metrics

• Daily active users
• Tasks completed autonomously
• User retention
• Average task completion rate
• User satisfaction score
• Memory retrieval accuracy
• Planning success rate

14. Technical KPIs

• Response latency < 2 sec
• Memory retrieval accuracy > 90%
• Tool execution success > 95%
• Planning success > 85%
• Screen understanding accuracy > 90%